

CSS Locators in Playwright

What are CSS Locators?

CSS locators are used to **find elements on a web page using CSS selector syntax**.

In Playwright, you can use them like this:

```
await page.locator('css_selector_here');
```

Common CSS Selector Patterns

1. Element Selector (tagname)

What it does:

Selects elements using their HTML tag name.

Example:

```
<button>Click Me</button>
```

```
await page.locator('button').click();
```

When to use:

- When the tag is unique on the page
- Not recommended if multiple same tags exist

2. ID Selector (#id)

What it does:

Selects an element using its unique id attribute.

Example:

```
<input id="username" />
```

```
await page.locator('#username').fill('admin');
```

When to use:

- Best when ID is unique
- One of the most reliable selectors

3. Class Selector (.classname)

What it does:

Selects elements using class name.

Example:

```
<button class="login-btn">Login</button>
```

```
await page.locator('.login-btn').click();
```

When to use:

- When class is stable
- Avoid if class names change frequently

4. Attribute Selector [attribute='value']

What it does:

Selects elements based on attribute and its exact value.

Example:

```
<input type="email" />
```

```
await page.locator("[type='email']").fill('test@mail.com');
```

When to use:

- When attributes are unique and stable

5. Multiple Attributes

What it does:

Selects elements that match multiple attributes.

Example:

```
<input type="text" name="username" />
```

```
await page.locator("input[type='text'][name='username']").fill('admin');
```

When to use:

- When one attribute is not enough

- Improves accuracy

6. Contains (Substring) [attribute*='partial']

What it does:

Selects elements where attribute contains a partial value.

Example:

```
<input id="user_123" />
```

```
await page.locator("[id*='user']").fill('admin');
```

When to use:

- When attribute values are dynamic
- Only part of value is known

7. Starts With [attribute^='start']

What it does:

Selects elements where attribute starts with a value.

Example:

```
<input id="user_123" />
```

```
await page.locator("[id^='user']").fill('admin');
```

When to use:

- When prefix is constant

8. Ends With [attribute\$='end']

What it does:

Selects elements where attribute ends with a value.

Example:

```
<input id="123_user" />
```

```
await page.locator("[id$='user']").fill('admin');
```

When to use:

- When suffix is constant

9. Child Selector (parent > child)

What it does:

Selects **direct child elements only**

Example:

```
<div class="form">  
  <input type="text" />  
</div>
```

```
await page.locator('.form > input').fill('admin');
```

When to use:

- When element is a direct child
- More specific than descendant selector

10. Descendant Selector (parent child)

What it does:

Selects **any nested child (not just direct)**

Example:

```
<div class="form">  
  <div>  
    <input type="text" />  
  </div>  
</div>
```

```
await page.locator('.form input').fill('admin');
```

When to use:

- When element can be anywhere inside parent

Summary

- tagname → Select by element type
- #id → Select by unique ID
- .classname → Select by class
- [attribute='value'] → Exact match
- [att1][att2] → Multiple conditions

- [attribute*='value'] → Contains
- [attribute^='value'] → Starts with
- [attribute\$='value'] → Ends with
- parent > child → Direct child
- parent child → Any descendant

Tips

- Prefer **ID and stable attributes** whenever possible
- Avoid long and complex selectors
- Use **multiple attributes** for better accuracy
- Be careful with dynamic values (use contains/starts-with)